Yuhao Shen

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Research Interests

My research focuses on **AI for Healthcare**, with a long-term vision of building trustworthy medical AI systems. I develop **medical LLMs and agents** for clinical support, including applications in dermatology. My work on **trustworthy medical AI** explores evidence-based and verifiable clinical reasoning, including approaches grounded in **clinical guidelines**. In parallel, I build **multimodal foundation models** to learn unified representations across medical images, text, and clinical data.

Education

The Chinese University of Hong Kong, Shenzhen — Shenzhen, China
Master of Philosophy in Computer Science — Sept. 2025 – Jun. 2027

- Supervisor: **Prof. Juexiao Zhou** | GPA: **3.6/4.0** | DYP Award

Hangzhou Dianzi University — Hangzhou, China
Bachelor of Engineering in Computer Science and Technology — Sept. 2021 – Jun. 2025

- Supervisor: **Prof. Renshu Gu** | GPA: **3.8/4.0** | Outstanding Graduate

Publications [Google Scholar] **Equal Contribution*

Journals

[J1] **Yuhao Shen***, Zhangtianyi Chen*, Yuanhao He*, Yan Xu, Shuping Zhang, Liyuan Sun, Zijian Wang, Yinghao Zhu, Yuyuan Yang, et al. *Trustworthy and Fair SkinGPT-R1 for Democratizing Dermatological Reasoning across Diverse Ethnicities*. **Under Revision at *Nature Communications***. [paper] [code]

[J2] Zhangtianyi Chen*, **Yuhao Shen***, Florensia Widjaja*, Yan Xu, Liyuan Sun, Zijian Wang, Hongyi Chen, Wufei Dai, et al. *SkinGPT-X: A Self-Evolving Collaborative Multi-Agent System for Transparent and Trustworthy Dermatological Diagnosis*. **Under peer review at *Nature Communications***. [paper] [code]

[J3] Ce Shen, Shuiming Wang, **Yuhao Shen**, et al. *Application of Machine-Vision-Based 3D Point Cloud Matching to Real-Time Detection of Bulk Materials*. *Automation Applications, 2023*. [paper]

Conferences

[C1] **Yuhao Shen***, Jiahe Qian*, Shuping Zhang, Zhangtianyi Chen, Tao Lu, Juexiao Zhou. *Towards Trustworthy Dermatology MLLMs: A Benchmark and Multimodal Evaluator for Diagnostic Narratives*. The 19th European Conference on Computer Vision (**ECCV 2026**). [paper] [code]

[C2] **Yuhao Shen***, Lang Cao*, Simo Du, Yuqing Wang, Juexiao Zhou, Hao Peng, Yue Guo. *MedGuideX: Internalizing Decision Logic from Clinical Guidelines into Large Language Models for Reliable Diagnostic Reasoning*. **Under Review at ICLR 2027**. [paper]

[C3] **Yuhao Shen**, Liyuan Sun, Yan Xu, Wenbin Liu, et al. *SkinCaRe: A Multimodal Dermatology Dataset Annotated with Medical Caption and Chain-of-Thought Reasoning*. The 3rd International Workshop on Multimedia Computing for Health and Medicine (**ACM MM 2026 Workshop**). [paper] [dataset]

[C4] **Yuhao Shen**, Xiyan Huang. *A Comparative Visualization Analysis of Neural Network Models using Grad-CAM*. Conference Covering Computer Science and Information Technology (**CSI 2025**). [paper]

[C5] **Yuhao Shen**. *Exploring Generalization Capability of U-Net Architecture through Domain Adaptation*. International Conference on Artificial Intelligence and Communication (**ICAIC 2024**). [paper]

[C6] Lang Cao, **Yuhao Shen**, Tianyang Luo, Yue Guo. *GuideSkill: Evolving Executable LLM Agent Skills for Guideline-Grounded Clinical Reasoning*. **Under Review at AAAI 2027**. [paper]

[C7] Jiahe Qian*, **Yuhao Shen***, Z. Chen, Juexiao Zhou, P. Wang. *CoTBox-TTT: Grounding Medical VQA with Visual Chain-of-Thought Boxes During Test-time Training*. **Under Review at ICLR 2027**. [paper]

[C8] Lang Cao, Binghang Lu, **Yuhao Shen**, Yue Guo. *MedRouter: Demystifying Knowledge Differences Across Medical LLMs for Routing-Based Reasoning*. **Under Review at ICLR 2027**. [paper]

[C9] Zhangtianyi Chen, Florensia Widjaja, Wufei Dai, Xiangjun Zhang, **Yuhao Shen**, Juexiao Zhou. *Beyond Prompt-Based Planning: MCP-Native Graph Planning-based Biomedical Agent System*. **Under Review at NAACL 2027**. [paper]

Technical Reports

[T1] **Ant Group** | Core Contributor
InclusionMed: Benchmarking Medical AI for Inclusive Healthcare. [code] [web]

Patents

[P1] *Machine Vision Universal Image Browser System*. 2023SR1244870.

[P2] *A Dock Production Management System Based on Big Data*. CN115936373A.

[P3] *A Laser Detection Method for Port Machine Equipment*. CN117970367A.

[P4] *An Automated Operation Method for Ship Hold Cleaning*. CN119329709A.

EXPERIENCES

Ant Healthcare, Ant Group Aug. 2026 – Present
Research Intern | Mentor: **Dr. Yuanfeng Ji**
Research: Guideline-driven Medical AI and Search Agent
Selected works: ***InclusionMed*** [T1], ***GuideEvo-Bench***

University of Illinois Urbana-Champaign (UIUC) Feb. 2026 – Jun. 2026
Research Intern | Supervisor: **Prof. Yue Guo**
Research: Guideline-Grounded Trustworthy Medical Reasoning
Selected works: ***MedGuideX*** [C2], ***GuideSkill*** [C6], ***MedRouter*** [C8]

The Chinese University of Hong Kong, Shenzhen Mar. 2025 – Aug. 2026
Research Assistant | Supervisor: **Prof. Juexiao Zhou**
Research: Multimodal LLMs & Agentic Systems for Dermatology
Selected works: ***DermBench*** [C1], ***SkinGPT-R1*** [J1], ***SkinGPT-X*** [J2], ***SkinCaRe*** [C3]

Xidian University Jul. 2023 – Sept. 2023
Research Intern | Supervisor: **Prof. Chao Yang**
Research: 3D Point-Cloud Perception | Selected work: ***3D Point Cloud Matching*** [J3]